ASSIGNMENT # 2

33. PSO

Group Members

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Task (a) - Part 1: Function Point Analysis

2.1 External Inputs (EI)

Req. ID	EI Name	Data Elements (DETs)	RETs	DET Count	Complexity	FP
FR-01	Customer Registration	CNIC, Name, Phone, Vehicle Number, Customer Type (Fleet/Individual)	1 (Customer Profile RET)	5	Low	3
FR-02	POS Offline Transaction Queuing	Liters purchased, Fuel type, Amount, POS ID, Timestamp, Sync Flag	2 (Transaction RET + Sync Queue RET)	6	Average	4
FR-03	Batch Data Sync	Batch ID, Transaction List, Total Checksum, Station ID, Sync Status	2 (Batch RET + Station RET)	5	Average	4
FR-04	Points Calculation & Awarding	Purchase amount, Multiplier, Points Awarded, Customer ID, New Balance	2 (Customer Profile + Loyalty Tier RETs)	5	Average	4
FR-05	Point Redemption Authorization	OTP, Redemption amount, POS ID, Customer ID, Updated Balance	1 (Redemption Log RET)	5	Low	3

EI Total: (2 Low × 3) + (3 Average × 4) = 6 + 12 = **18 FP**

2.2 External Outputs (EO)

Req. ID	EO Name	Data Elements (DETs)	RETs	DET Count	Complexity	FP
FR-06	Transaction SMS Notification	Customer Name, Points Awarded/Redeemed, Current Balance, Station Location	2 (Customer Profile + Transaction RET)	4	Low	4
FR-07	Fleet Manager Analytics Dashboard	Total fuel consumed, Points earned, Vehicle breakdown, Station heatmaps, Date range	3 (Customer RET + Vehicle RET + Transaction RET)	5	Average	5

EO Total: (1 Low × 4) + (1 Average × 5) = 4 + 5 = **9 FP**

2.3 Internal Logical Files (ILF)

Req. ID	ILF Name	Record Types (RETs)	RET Count	DET Description	Complexity	FP
FR-08	Unified Customer Database	Customer Profile, Vehicle Log, Transaction History, Points Balance	4	Name, CNIC, Phone, Vehicle Make, Reg Number, Liters, Amount, Points, Tier Status, Registration Date (15+ DETs)	Average	10

ILF Total: (1 Average × 10) = **10 FP**

2.4 External Interface Files (EIF)

Req. ID	EIF Name	External Source	RET Count	DET Description	Complexity	FP
FR-09	SAP/ERP Financial Ledger	PSO Legacy SAP ERP	2 (General Ledger RET + Point	Account ID, Daily Point Liability, Monetary Value, Station Reconciliation,	Low	5

Req. ID	EIF Name	External Source	RET Count	DET Description	Complexity	FP
			Liability RET)	Sync Timestamp (5 DETs)		

EIF Total: (1 Low × 5) = **5 FP**

2.5 External Inquiries (EQ)

Req. ID	EQ Name	Files Referenced	RET Count	DET Description	Complexity	FP
FR-10	Real-time Balance Inquiry	Unified Customer Database	1	Customer ID, CNIC, Current Points, Equivalent Rupee Value (4 DETs)	Low	3

EQ Total: (1 Low × 3) = **3 FP**

2.6 UFP Summary

FP Component	Requirements Mapped	Complexity Calculation	Count	FP Count
External Inputs (EI)	FR-01, FR-02, FR-03, FR-04, FR-05	Low: 2 × 3 = 6 Avg: 3 × 4 = 12	5	18
External Outputs (EO)	FR-06, FR-07	Low: 1 × 4 = 4 Avg: 1 × 5 = 5	2	9
Internal Logical Files (ILF)	FR-08	Avg: 1 × 10 = 10	1	10

FP Component	Requirements Mapped	Complexity Calculation	Count	FP Count
External Interface Files (EIF)	FR-09	Low: 1 × 5 = 5	1	5
External Inquiries (EQ)	FR-10	Low: 1 × 3 = 3	1	3
TOTAL UFP			**10**	**45**

Task (a) - Part 2: Functional and Non-Functional Requirements

10 Functional Requirements

#	Requirement	FP Type	Description
FR-01	Customer Registration	EI	Fuel station managers register new profiles via POS using CNIC, phone number, and vehicle details.
FR-02	POS Offline Transaction Queuing	EI	If internet is lost, the local POS system logs the loyalty transaction locally, flagging it for future sync.
FR-03	Batch Data Sync	EI	The local POS automatically pushes all queued offline transactions to the central server when network is restored.
FR-04	Points Calculation & Awarding	EI	The Loyalty Engine calculates DigiCash points based on liters purchased and updates the customer's balance.
FR-05	Point Redemption Authorization	EI	The system verifies OTP and point balance to authorize a discount on the current fuel purchase.

#	Requirement	FP Type	Description
FR-06	Transaction SMS Notification	EO	The CRM automatically triggers an SMS via a third-party gateway detailing points earned/spent and current balance.
FR-07	Fleet Manager Analytics Dashboard	EO	Corporate clients can log in to view aggregated data on fuel consumption and points across their registered vehicle fleet.
FR-08	Unified Customer Database	ILF	The central repository storing all demographic data, vehicle logs, tier statuses, and transaction histories.
FR-09	SAP/ERP Financial Ledger	EIF	The CRM reads/writes aggregated financial point liabilities to PSO's existing SAP backend to ensure accounting accuracy.
FR-10	Real-time Balance Inquiry	EQ	Station attendants can query a customer's CNIC to view their current point balance without executing a transaction.

10 Non-Functional Requirements

#	Requirement	Category	Description
NFR-01	Offline Availability	Reliability	The POS module must retain up to 72 hours of offline transaction data without loss during prolonged outages.
NFR-02	Scalability	Scalability	Must support concurrent data synchronization from up to 3,500+ retail outlets nationwide.
NFR-03	Transaction Performance	Performance	Point redemption queries must resolve in < 2.5 seconds to prevent bottlenecks at fuel pumps.

#	Requirement	Category	Description
NFR-04	PII Security	Security	All CNIC and customer data must be encrypted at rest and in transit using AES-256 encryption.
NFR-05	Legacy Hardware Compatibility	Usability	The local POS sync software must be lightweight enough to run on legacy Pentium-era hardware at rural stations.
NFR-06	Regulatory Compliance	Compliance	System architecture and data storage policies must comply with State Bank of Pakistan (SBP) data privacy laws.
NFR-07	ERP Integration Uptime	Reliability	The SAP integration bridge must maintain 99.9% uptime for daily financial reconciliations.
NFR-08	Bilingual Interface	Usability	Customer-facing SMS alerts and printed POS receipts must support both Urdu and English text.
NFR-09	Modular Updatability	Maintainability	The Loyalty Engine can be updated independently without taking the main Customer Database offline.
NFR-10	Low-Bandwidth Sync	Performance	Data sync payloads must be compressed (<20KB per transaction) to work effectively over 2G/3G networks in remote areas.

Task (b) - COCOMO-II Early Design Modeling

1) Estimation of KLOC:

- **Total UFP:** 45
- **Language Factor:** The project involves a mix of legacy integration (C/C++) and modern backend/mobile (Java/C#). We use the industry-average conversion factor of 50 LOC per Function Point.
- **Calculation:** 45 × 50 = 2250 SLOC
- **Conversion to KLOC:** 2250 / 1000 = **2.25 KLOC**

2) Scale Factors (SF):

Scale Factor	Rating	Value	Reasoning & Context
PREC (Precedentedness)	Nominal	3.72	PSO operates massive logistics, but a centralized offline-first CRM interacting directly with end-consumers is a relatively new endeavor for them.
FLEX (Development Flexibility)	Low	4.05	High constraint. The system must rigidly adhere to SBP data regulations and predefined legacy SAP financial ledger formats.
RESL (Architecture / Risk Resolution)	High	2.83	As noted in Assignment 1, a detailed Risk Register has been developed, and connectivity risks are mitigated via the offline-first architecture.
TEAM (Team Cohesion)	Nominal	3.29	Mixed team of legacy IT staff, new mobile developers, and retail operations managers requiring heavy coordination.
PMAT (Process Maturity)	Nominal	4.68	Standard enterprise IT PMO standards apply.

- **SF Calculation:**

  B = 0.91 + 0.01 x (3.72 + 4.05 + 2.83 + 3.29 + 4.68)

  B = 0.91 + 0.1857 = 1.0957

3) Effort Multipliers (EM):

Effort Multiplier	Rating	Value	Reasoning & Context
RCPX (Product Reliability & Complexity)	High	1.33	System handles direct financial liabilities (DigiCash) and complex offline queuing (FR-02) across thousands of nodes.

Effort Multiplier	Rating	Value	Reasoning & Context
RUSE (Developed for Reusability)	Nominal	1.00	Developed specifically for PSO's proprietary hardware and SAP setup; not intended for external resale.
PDIF (Platform Difficulty)	High	1.29	Must interface with highly constrained legacy POS hardware (NFR-05) and unstable 2G networks (NFR-10).
PERS (Personnel Capability)	Nominal	1.00	Standard, capable enterprise development team.
PREL (Personnel Continuity)	Nominal	1.00	Standard corporate retention expected throughout the 1-year development cycle.
FCIL (Facilities)	Nominal	1.00	Standard PSO corporate IT infrastructure and environments.
SCED (Required Schedule)	Nominal	1.00	Standard 12-month rollout as defined in the Assignment 1 business case.

- **EAF Calculation:**

  EAF = 1.33 x 1.00 x 1.29 x 1.00 x 1.00 x 1.00 x 1.00 = 1.7157

4) Final Effort Estimation:

- **Formula:** Effort = A x (KLOC)^B x EAF (Where A = 2.94 as per COCOMO II)
- **Calculations:**

  $(2.25)^{1.0957}$= 2.44

  Effort = 2.94 x 2.44 x 1.7157

  Effort = 12.30 Person-Months

5) Development Duration (Assuming a core team of 4 dedicated developers/analysts):

- **Dev. Duration** = 12.30 / 4 = **3.07 Months**